

Joseph CLARKE

I am a Postdoctoral Research Associate and climate scientist at the University of Exeter. I use the full model hierarchy to study tipping in the climate-carbon system. For example, I have analysed Amazon dieback in CMIP6 data, I have run experiments using the complex land-surface model JULES, and I have designed conceptual models of heat transport in soils to study the compost bomb effect. In addition to my background in climate science, I use techniques from dynamical systems, statistical and stochastic dynamics to understand nonlinear phenomena in the Earth system. I am interested in how early warning signals can be used to help avoid tipping points, particularly using high-resolution Earth observation data.

PERSONAL INFORMATION

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EDUCATION

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| 2019–2023 | PhD in Mathematics, University of Exeter ,
PHD THESIS: ‘Tipping Points and Early Warning Signals in the Climate-Carbon System’, supervised by Professor Peter Cox and Professor Peter Ashwin. |
| 2015–2019 | MPhys Physics (1 st Class), Brasenose College, Oxford University ,
SPECIALISING IN: The Physics of Atmospheres and Oceans and Condensed Matter Physics
MASTER’S THESIS: ‘Heat-engine based representation of convection with phase change in exoplanet atmospheres’ with Professor Raymond Pierrehumbert. |

EMPLOYMENT AND WORK EXPERIENCE

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| 2023– | Postdoctoral Research Associate
At the university of Exeter, I investigate early warning signals and the stability of the climate-carbon system |
| 2019–2023 | Postgraduate Teaching Assistant.
Assisted teaching in the following modules: ‘Probability, Statistics and Data’, ‘Mathematics of Climate Change’, ‘Ordinary Differential Equations’, ‘Partial Differential Equations’, ‘Fluid Dynamics’ |
| 2018 | Summer Project with Dr Hannah Christensen, University of Oxford.
Took a dynamical systems approach to constraining climate sensitivity. I worked with Linear Inverse Models, Empirical Orthogonal Functions, autoregressive models, Python, the Fluctuation Dissipation Theorem and undertook analysis of CMIP data |
| 2017 | Summer Intern at the Earth Observation Data Group, University of Oxford.
Developed a website to display observations of atmospheric anomalies in near real time. This involved processing the large amount of data generated and converting it to an understandable form. |

POSITIONS OF RESPONSIBILITY

2025 –	XCS Seminar Organiser I run the XCS (Exeter Climate Systems) seminars.
2025	Global Tipping Points Conference Committee Organising the Global Tipping Points 2025 conference, an interdisciplinary conference with over 500 attendees.
2024–2025	GSI Seminar Organisation Committee We run the Global Systems Institute seminar series. We invite speakers, organise rooms as well as provide AV support. Events include traditional seminars, assemblies and film screenings.
2024	Pint of Science Organiser Outreach activity arranging science talks for the public. Involved working with speakers, operating AV equipment and running the events.
2017–2019	Web Officer, Physics Joint Consultative Committee (PJCC) Responsible for maintaining the PJCC website using the Django framework. Liaising with the physics department to represent student interests
2016	Producer and Sound Technician, The Importance of Being Ernest, Brasenose Arts Week Responsible for operating microphones and playing music during performance. Designed and produced posters as well as acquired props. Assisted with rehearsals

RESEARCH SKILLS

Scientific programming using Python, Fortran and C. Linux computer systems including the JASMIN platform. Bifurcation and rate induced tipping. Linear Response. Spectral analysis. CMIP6 data analysis. Development and running of land surface models (JULES). Analysis of conceptual models with dynamical systems techniques. Fluctuation Dissipation Theorem. Emergent Constraints. Carbon Cycle biogeochemistry.

PUBLICATIONS

- 2026 | Abrams, Jesse F., Laura M. Pereira, Viktoria Spaiser, Nico Wunderling, Sebastian Villasante, J. David Tàbara, Reinette Biggs, David I. Armstrong McKay, Caroline Zimm, Manjana Milkoreit, Sara M. Constantino, Ramit Debnath, Vasilis Dakos, Paul D. L. Ritchie, Steven R. Smith, Juan C. Rocha, Sonia Kéfi, Joshua E. Buxton, Chris A. Boulton, Kai Greenlees, Sirkku Juhola, Avit Bhowmik, Lukas P. Fesenfeld, **Joseph Clarke**, James G. Dyke, Laurie Laybourn, Steven J. Lade, Mia Marjanovic-Abrams, Ben Dickenson Bampton, and Timothy M. Lenton (2026). “Integrating Tipping Point Concepts Across Diverse Systems”. In: *Communications Sustainability* 1.1, p. 68. ISSN: 3059-4308. DOI: [10.1038/s44458-026-00063-5](https://doi.org/10.1038/s44458-026-00063-5). URL: <https://doi.org/10.1038/s44458-026-00063-5>.
- | Boardman, P. J. L., **J. Clarke**, P. M. Cox, C. Huntingford, C. D. Jones, and M. S. Williamson (2026). “Emergent constraints on climate sensitivity and recent record-breaking warm years”. In: *Earth System Dynamics* 17.3, pp. 829–841. DOI: [10.5194/esd-17-829-2026](https://doi.org/10.5194/esd-17-829-2026). URL: <https://esd.copernicus.org/articles/17/829/2026/>.
- | **Clarke, Joseph**, Chris Huntingford, Paul D L Ritchie, and Peter Cox (May 2026). “Conditions for skilful spatial and temporal tipping point early warning signals”. In: *Journal of Physics: Complexity* 7.2, p. 025007. DOI: [10.1088/2632-072X/ae6217](https://doi.org/10.1088/2632-072X/ae6217). URL: <https://doi.org/10.1088/2632-072X/ae6217>.

- Ritchie, Paul D L, Norman J Steinert, Jesse F Abrams, Hassan Alkhayuon, Constantin W Arnscheidt, Nils Bochow, Ruth R Chapman, **Joseph Clarke**, Donovan P Dennis, Jonathan F Donges, Bernardo M Flores, Julius Garbe, Annika Högner, Chris Huntingford, Timothy M Lenton, Johannes Lohmann, Kerstin Lux-Gottschalk, Manjana Milkoreit, Tessa Möller, Paul Pearce-Kelly, Laura Pereira, Courtney Quinn, Carl-Friedrich Schleussner, Simone M Stuenzi, Didier Swingedouw, Larissa N Van der Laan, Kirsten Zickfeld, and Nico Wunderling (2026). "The implications of overshooting 1.5 °C on Earth system tipping elements - a review". In: *Environmental Research Letters* 21.4, p. 043001. DOI: [10.1088/1748-9326/ae3cad](https://doi.org/10.1088/1748-9326/ae3cad). URL: <https://doi.org/10.1088/1748-9326/ae3cad>.
- 2025 | **Clarke, J.**, C. Huntingford, P. D. L. Ritchie, R. Varney, M. S. Williamson, and P. Cox (2025). "Conditions for Instability in the Climate-Carbon Cycle System". In: *Earth System Dynamics* 16.6, pp. 2087–2100. DOI: [10.5194/esd-16-2087-2025](https://doi.org/10.5194/esd-16-2087-2025). URL: <https://esd.copernicus.org/articles/16/2087/2025/>.
Highlight paper. I found the conditions under which the pre-industrial climate-carbon cycle system would be linearly unstable.
- 2024 | **Clarke, J** (2024). "Tipping Points and Early Warning Signals in the Climate-Carbon System". PhD thesis. University of Exeter. URL: <https://hdl.handle.net/10871/134880>.
My thesis which examined tipping points in the carbon cycle and investigated early warning signals.
- Huntingford, Chris, Peter M Cox, Paul DL Ritchie, **Joseph J Clarke**, Isobel M Parry, and Mark S Williamson (2024). "Acceleration of daily land temperature extremes and correlations with surface energy fluxes". In: *npj Climate and Atmospheric Science* 7.1, p. 84. URL: <https://doi.org/10.1038/s41612-024-00626-0>.
- 2023 | **Clarke, Joseph J**, Chris Huntingford, Paul DL Ritchie, and Peter M Cox (2023). "Seeking more robust early warning signals for climate tipping points: the ratio of spectra method (ROSA)". In: *Environmental Research Letters* 18.3, p. 035006. URL: <https://doi.org/10.1088/1748-9326/acbc8d>.
Conventional early warning signals make assumptions about the noise the system experiences, which can make them ineffective when those assumptions are not satisfied. I introduced a new early warning signal which still works well for more general, time correlated noise. I applied this to Amazon dieback in CMIP6.
- 2022 | Huntingford, Chris, Peter M Cox, Mark S Williamson, **Joseph J Clarke**, and Paul DL Ritchie (2022). "Emergent constraints for the climate system as effective parameters of bulk differential equations". In: *Earth System Dynamics Discussions* 2022, pp. 1–15. URL: <https://doi.org/10.5194/esd-14-433-2023>.
- Ritchie, Paul DL, Isobel Parry, **Joseph J Clarke**, Chris Huntingford, and Peter M Cox (2022). "Increases in the temperature seasonal cycle indicate long-term drying trends in Amazonia". In: *Communications Earth & Environment* 3.1, pp. 1–8. URL: <https://doi.org/10.1038/s43247-022-00528-0>.
- 2021 | **Clarke, Joseph**, Chris Huntingford, Paul Ritchie, and Peter Cox (2021). "The compost bomb instability in the continuum limit". In: *The European Physical Journal Special Topics*, pp. 1–7. URL: <https://doi.org/10.1140/epjs/s11734-021-00013-3>.
I derived a conceptual model for the compost bomb, showing that the diffusion of heat does not suppress the compost bomb instability. I demonstrated that compost bombs could be caused by hot summers and not just rapid global warming.

Ritchie, Paul DL, **Joseph J Clarke**, Peter M Cox, and Chris Huntingford (2021). "Overshooting tipping point thresholds in a changing climate". In: *Nature* 592.7855, pp. 517–523. URL: <https://doi.org/10.1038/s41586-021-03263-2>.

TALKS AND RESEARCH SESSIONS

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| 2026 | <p>Conditions for skilful spatial and temporal tipping point early warning signals, EXtremes and Rare Events in Climate and related applications (ExREC)</p> <p>Invited Speaker</p> |
| 2025 | <p>Conditions for Instability in the Climate-Carbon System, Exeter Tipping Points Seminar</p> <p>Gave this talk to the Exeter Tipping Points seminar, an inter-disciplinary audience</p> <p>Spatial Early Warning for Rapidly Forced Systems, The Mathematics of Climate Tipping Points and their Impacts</p> <p>Mathematics workshop on tipping points</p> <p>Tipping Points in the Biosphere, Global Tipping Points Conference</p> <p>I helped run this research session</p> |
| 2024 | <p>Conditions for stability in the Carbon-Climate System, Dynamics Days Europe 2024</p> <p>Invited to present my work on bifurcations in the climate-carbon system</p> |
| 2023 | <p>Spatial Early Warning Signals for Rapidly Forced Systems, EGU 2023</p> <p>Presented my work on how spatial information can be helpful for giving early warning signals when the system is forced quickly compared to its own timescale</p> |
| 2022 | <p>Early Warning Signals for Tipping Points, XCS Seminar, University of Exeter</p> <p>Presented my research to a more general audience on early warning signals, looking at time correlated noise and spatial early warning signals</p> <p>Early Warning Signals: Beyond White Noise, EGU 2022</p> <p>Presented my work on early warning signals with time correlated noise and early warning signals for Amazon dieback to the Nonlinear Processes Division</p> |
| 2021 | <p>Early Warning Signals: Beyond White Noise, Tipping Points Seminar, Potsdam Institute for Climate Impact Research</p> <p>Invited to present on early warning signals for systems subject to time correlated noise</p> <p>The Compost Bomb in the Continuum Limit, Dynamics Internal Seminar, University of Exeter</p> <p>Discussed rate and bifurcation induced tipping for the simple compost bomb system I had derived for a mathematical audience</p> |
| 2020 | <p>The Compost Bomb in the Continuum Limit, Emergent Constraints and Tipping Points Workshop</p> <p>Took part in workshop developing connections between tipping points and emergent constraints</p> |

REFEREES

PROFESSOR PETER COX: p.m.cox@exeter.ac.uk
PROFESSOR PETER ASHWIN: p.ashwin.exeter.ac.uk
DR PAUL RITCHIE: paul.ritchie.exeter.ac.uk